

Exercise 1 – LTspice

Use the file on Brightspace: **Week4_Ex1.asc**. It contains the schematic of a pseudo-differential pair and of a differential pair.

- A. Plot the small-signal differential gain while sweeping the input CM from 0 V to 1.8 V. Explain the behaviour.

You can use `.step VCM 0 1.8 10m`
 `.ac list 1`

- B. Copy the diff. pair circuit and replace the ideal current source with a current mirror. Repeat point A for the new circuit. Explain the differences of the new circuit.
- C. Simulate the CM gain for the 3 circuits. What is the CMRR?
- D. Add a mismatch of 1% in the load resistors. Plot A_{CM-DM} for the 3 circuits.

Exercise 2 – LTspice

Use the file on Brightspace: **Week4_Ex2.asc**. It contains the schematic of a 5-transistor OTA.

- A. Find the small-signal DM gain both by AC analysis and by sweeping the DM input.
- B. Plot the small-signal differential gain while sweeping the input CM from 0 V to 1.8 V. Explain the behaviour.

You can use `.step VCM 0 1.8 10m`
 `.ac list 1`

- C. Simulate the CM gain. What is the CMRR?
- D. Connect the output to the negative input, i.e. connect the Ota as a buffer. Add an AC excitation on the positive input. Plot the gain as a function of the DC value on the positive input.